

Course Syllabus

SOFT 468/CSCE 468/ CSCE 868 – Requirements Elicitation, Modeling, and Analysis

Spring 2026

School of Computing

University of Nebraska - Lincoln

Instructor : Dr. Bonita Sharif

Class Number: CSCE-468-001, SOFT-468-001 (undergrad),
CSCE-868-001 (graduate)

Class Meetings : T R 12:30 – 1:45 pm

Classroom: 105 TEAC

Office : 103 Schorr

Office hours : Email for appointment (in person or Zoom call)

Phone: (402) 472-5084

Email : bsharif@unl.edu (<mailto:bsharif@unl.edu>)

Homepage : <https://www.shbonita.me>  (<https://www.shbonita.me>)

Course Updates : The course will be conducted on Canvas([https://
canvas.unl.edu/](https://canvas.unl.edu/)) for schedule, lectures, and assignments.

Course Description

In-depth coverage of processes, methods and techniques for determining, or deciding, what a proposed software system should do. Topics include the requirements engineering process, identification of stakeholders, requirements elicitation techniques, methods for informal and formal requirements documentation, techniques for analyzing requirements models for consistency and completeness, and traceability of requirements across system development and evolution. Tool support for modeling functional and non-functional requirements to support elicitation and analysis will be studied.

TA/Grader:

Kang-il Park - kangil.park@huskers.unl.edu (<mailto:kangil.park@huskers.unl.edu>)

Prerequisites:

By Course:

- Successful completion of SOFT 261 or CSCE 361.

By Topic:

- Mastery of high-level programming languages and basic data structures and algorithms, and propositional logic.
- Familiarity with fundamental software engineering processes and techniques.

Credit Hours:

3

Textbook

There is no official textbook for the course. Canvas will provide all the necessary materials needed.

Reference Textbooks (optional readings)

- “Requirements Engineering: From System Goals to UML Models to Software Specifications”, Axel van Lamsweerde, Wiley, 2009
- “User Story Mapping”, Jeff Patton, O’Reilly Media, 2014
- “Requirements Engineering Fundamentals, Principles and Techniques”, Klaus Pohl, Springer, 2025, 2nd Edition

Course Goals

1. Study and apply the foundational concepts of requirements engineering.
2. Emphasize the role of system goals in shaping requirements engineering activities.
3. Methods for helping stakeholders communicate system goals clearly and completely.
4. Explore the use of structural and behavioral specifications of system requirements.
5. Exposure to a wide-range of functional and non-functional requirements categories.
6. Methods for integrating requirements engineering into agile development workflows.
7. Learn to design for security. Security is addressed in terms of requirements elicitation (e.g., secure data transmission. secure data storage requirements) and through concepts such as information hiding and design.

Conducting the Class

The class will be structured as a series of weekly modules. Each module will involve student reading, instructor lecturing, live group work sessions in class, and an exercise or project component that allows students to learn the module content by “doing”. Prior to each class meeting, links will be provided when necessary for reading the assigned content. There will be ample opportunity for interactive discussions. Students are expected to actively participate in these discussions.

AI Disclosure

If you are using AI tools, please copy your prompt and/or process into the generated solution as a comment (or have your agent do that):

1. You wrote a precise requirement spec - why throw it away?
2. It will help the next person understand the artifact/code.
3. It is a good context for the AI when you revise the artifact, too.
4. It will help researchers who want to study these tools' accuracy and effectiveness.
5. It is a good way to label artifacts as “AI-generated” or “AI-assisted”.

Grading

Readings and Reflections

You will be assigned reading material and asked to complete a reading reflection based on several framework options provided. This will be assigned on a weekly basis. There are eight choices in the framework and you will be expected to complete 5 of the 8 by the end of the semester.

Assignments

Grading in this course will combine assessments of individual and group activities. Individual activities will be assigned as assignments on Canvas. Group assignments will also be assigned on Canvas and may be conducted both in class and/or expected to be completed outside of class in respective groups.

Midterm Exam

There will be one midterm exam in this course. The midterm is a closed-book comprehensive assessment over the course material up to the point of the exam. It will be held in class on paper.

The midterm exam will be held on March 26th. An in-class exam review will be held on March 24th. This is the week after you get back from Spring Break.

Group Project

Students will work in a group to engage in a course project that develops the requirements for a realistic software system. Student teams will propose the project, provide a midterm project update (Weeks 8 and 9), and then prepare and present their final system requirements and deliverables (Week 15 onwards). The project will require you to keep running totals of your time spent. You will also be required to peer review your peers' midterm projects and presentations. At the end of the project, you will also peer review your own team members.

There is no final exam. The project presentations will run into the **final exam slot (<https://registrar.unl.edu/calendars/final-exam-schedule/spring-final-exam-schedule/>)** for this class which is on **Friday, May 8 from 10:00 a.m. to 12:00 p.m.**

Graduate Students

Graduate students will, in addition to the above, select, study, and present a lecture on an emerging topic in requirements engineering. Students are encouraged to select a topic that supports their research or other academic work and will be presented with a set of suggested topics to consider. Graduate students will also have one additional assignment to critically review a technical paper. These two assignments are not available to undergraduate students.

The assignment percentages will be as follows:

Activity	Undergraduate %	Graduate %
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Readings and Reflections (Individual)	5	5
Class Participation (Exit Tickets/In-class polling/ Surveys)	5	5
Assignments (Individual and/ or Group)	30	20
Midterm Exam	20	20
Project (Group)	30	30
Final Project Presentation (Group)	10	10
Review Technical Paper		5
Paper Presentation		5

Grading Scale

Final grades will be assigned based on the following cutoff percentages:

A+ 97%

A 93%

A- 90%

B+ 87%

B 83%

B- 80%

C+ 77%

C 73%

C-	70%
D	60%
F	<60%

Course and Institutional Policies

Class Participation and Discussion

There will be plenty of opportunities for student discussion related to the topics covered during class meetings. It is expected that everyone will participate in discussions and activities. To discuss, you must attend. Attendance is necessary and expected.

Attendance Policy

Regular class attendance is necessary and expected. Regular class attendance, arriving to class on time, class participation, and attention will all be considered in computing the final course grade. To participate in class discussions, you must attend. If a student arrives late or leaves early during a lecture, it is the student's responsibility to contact the instructor to ensure that they have not been marked absent for that lecture. Exit tickets will be used to track attendance and will be distributed at the end of each class. You are allowed to miss 3 classes, and it will not count negatively towards your class participation grade.

Late Policies

To ensure that work can be graded and returned promptly, I strongly encourage you to submit everything on time. In general, you will have adequate time to complete each assignment. However, it is best to begin working on each assignment early so that you have plenty of time to complete it.

Makeup Policies

It is the student's responsibility to make up any missed material. Make-ups will only be given in the case of an excused absence or documented, valid emergency. If you know you will miss a class, please notify me in advance. This includes presentations and assignments. I encourage you to contact me as soon as possible if an emergency arises. In other cases, no makeup will be allowed for presentations and assignments.

Emergencies

I understand that these are unprecedented times. If you have an urgent situation that prevents you from completing your work on time, please don't hesitate to request accommodation. With advance notice, we can take significant steps to protect your grade from being affected by an emergency. You are expected to be proactive in this manner; last-minute pleas for accommodations are less likely to

be granted.

Academic Integrity

Academic honesty and personal integrity are the foundation upon which a quality education is built. To maintain high scholastic standards and to ensure each student the right to obtain a quality education, the University cannot tolerate academic dishonesty, e.g., cheating or plagiarism. You are required to adhere to the Department Policy and all University policies related to academic integrity. Please read the policy at https://computing.unl.edu/academic-integrity-policy_ (https://computing.unl.edu/academic-integrity-policy_) to which all students are held accountable. You are expected to read, understand, and follow this policy. Violations will be dealt with on a case-by-case basis and may result in a failing assignment or a failing grade for the course.

Copyright/Plagiarism

As commonly defined, plagiarism consists of passing off as one's own the ideas, words, writings, etc... which belong to another. In accordance with this definition, you are committing plagiarism if you copy the work of another person and turn it in as your own, even if you have the permission of that person. Plagiarism is one of the worst academic sins, for the plagiarist destroys the trust among colleagues without which research cannot be safely communicated.

You are responsible for developing and writing the ideas in your assignments, and they are expected to be your own original work. Copying content from a web site or other resources will be considered an act of plagiarism and will result in punitive actions within the range described in the University Student Rules.

Incomplete Grades

Incompletes will be only given if the student has already substantially completed the major requirements of the course. Refer to the university policy at <https://registrar.unl.edu/academic-standards/policies/incomplete-grade/> (<https://registrar.unl.edu/academic-standards/policies/incomplete-grade/>).

Email Checking Policy

It is the Department policy that all students in CSCE/SOFT courses are expected to regularly check their email so they do not miss important announcements. Make sure you check your email. I will periodically email the class as needed during the semester using the email you have on file.

Device Policy

We will use smartphones and laptops to facilitate activities and project work in class. However, research and student feedback clearly show that using devices for non-class-related activities not only harms your own learning, but also other students' learning as well. I only allow device usage during activities that require the use of devices. At all other times, please refrain from using your

device for things other than note-taking.

Communication Etiquette

Here's a general article on how to email your professors.

<https://www.insidehighered.com/views/2015/04/16/advice-students-so-they-dont-silly-emails-essay>.  (<https://www.insidehighered.com/views/2015/04/16/advice-students-so-they-dont-sound-silly-emails-essay>)

Student Expectations

- The student is responsible for devoting significant out-of-class time to complete all assignments using his/her own computer and/or lab computers. There will be cases where class time will be allocated for group discussions to initiate assignments and project deliverables.
- The lecture is the student's responsibility; if class is missed, it is in the student's best interests to get the notes from a fellow student, preferably one who takes good notes.
- The student is responsible for grasping all material and concepts presented in class, which may or may not originate from the assignments and material provided.
- Check Canvas often. It will contain up-to-date schedule information and any notes or assignments. Check Piazza often to keep up to date with team messages.
- Any modifications to the syllabus will be made on Canvas and schedule. It is the student's responsibility to keep up-to-date and adjust to the changes. Changes will also be announced in class.
- Students are expected to exhibit professional conduct that is respectful to other students. Each student is responsible for ensuring their own personal professional conduct and the professionalism of their work.
- A student who does not fully understand a lecture or assignment will assume the responsibility of raising his/her questions in class. If further explanation is required on an individual basis, the student is encouraged to schedule an appointment with the instructor outside of class hours.
- When working on group projects, each student is expected to carry his/her share of the load and to do their share of the work. Peer evaluations will be conducted to make sure each group member has a voice in the participation portion of the grade.
- All work needs to be original.
- Plan ahead. Don't put off working on the assignments/project until a day or two before they are due. Begin working on the assignments early, so you have plenty of time to complete them and clarify any questions you may have.
- Backup, backup, backup! A few minutes spent making a backup could save you hours if a computer crashes or if a disk becomes damaged. Take advantage of cloud services such as Box, Dropbox, EverNote, and other sky drives. Using a good version control system, such as GitHub, is also recommended. As a student, you can request to receive the educational plan that comes with free private repositories. <https://github.com/edu>  (<https://github.com/edu>)

- A reading list of chapters and papers will be posted. Readings are to be completed before the lecture so students come to class ready to ask questions and/or discuss the material.
- If you have any problems, including understanding the material that we cover in class, ask in class or please email me and/or bring your questions/problems to office hours.

Students with Disabilities

Students with disabilities are encouraged to contact the instructor for a confidential discussion of their individual academic accommodation needs. This includes students with mental health disabilities like depression and anxiety. It is the policy of the University of Nebraska-Lincoln to provide individualized accommodations to students with documented disabilities that may affect their ability to fully participate in course activities or to meet course requirements. To receive accommodation services, students must be registered with the Services for Students with Disabilities (SSD) office, 232 Canfield Administration, 472-3787.

Student Resource Center

The CSE Student Resource Center located in Avery Hall 12, is staffed by student tutors who are available to help you with your CS and CE courses. The staff can also provide technical support on issues such as problems logging in, printing, installing an app, etc., and point you in the right direction if you need additional help. For more information, go to <https://computing.unl.edu/current-undergraduate#SRC> (<https://computing.unl.edu/current-undergraduate#SRC>)

Writing Center

The writing center is a valuable resource to get feedback on your writing assignments if needed. Visit <https://www.unl.edu/writing/hours-locations> (<https://www.unl.edu/writing/hours-locations>) for more information. They are located in 102 Andrews Hall and other satellite locations.

Contact Form

The School of Computing has an anonymous contact form at <https://computing.unl.edu/anonymous-department-feedback-form> (<https://computing.unl.edu/anonymous-department-feedback-form>) that you may use to voice your concerns about any problems in the course or department if you do not wish to be identified.

Statement of Non-Discrimination

The University of Nebraska does not discriminate based on race, color, ethnicity, national origin, sex, pregnancy, sexual orientation, gender identity, religion, disability, age, genetic information, veteran status, marital status, and/or political affiliation in its programs, activities, or employment.

Counseling Services

UNL offers a variety of options to students to aid them in dealing with stress and

adversity. Counseling and Psychological Services (CAPS) - <https://caps.unl.edu/> (<https://caps.unl.edu/>) - is a multidisciplinary team of psychologists and counselors that works collaboratively with Nebraska students to help them explore their feelings and thoughts and learn helpful ways to improve their mental, psychological and emotional well-being when issues arise. CAPS can be reached by calling 402-472-7450.

Have you tried the Big Red Pawp Up where you pet a dog and de-stress?

Love Library South - Peterson Room (LLS 221) | Wednesdays 12-1:30pm

February 11th, March 11th, April 8th, April 29th

UNL Course Policies

Students are responsible for knowing the university policies and resources found on this page (<https://go.unl.edu/coursepolicies> (<https://go.unl.edu/coursepolicies>)):

- University-wide Attendance Policy
- Academic Integrity Policy
- Services for Students with Disabilities
- Mental Health and Well-Being Resources
- Final Exam Schedule
- Fifteenth Week Policy
- Emergency Procedures
- Nondiscrimination Policy
- Title IX Policy
- Other Relevant University-Wide Policies

Course Summary:

Date	Details	Due
Tue Jan 13, 2026	 Exit Ticket - Jan 13 (https://canvas.unl.edu/courses/209906/assignments/2266490)	due by 11:59pm
Thu Jan 15, 2026	 Background Questionnaire (https://canvas.unl.edu/courses/209906/assignments/2266299)	due by 11:59pm
Tue Jan 20, 2026	 Syllabus Quiz (https://canvas.unl.edu/courses/209906/)	due by 11:59pm

Date	Details	Due
	assignments/2266301	